

Study and Critical Analysis

ML-SemReg: Boosting Point Cloud Registration with Multi-level Semantic Consistency

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1. Introduction

Point cloud registration is a fundamental task in computer vision and 3D perception, aiming to align two or more point clouds into a common coordinate system. It is a critical component in many real-world applications such as autonomous driving, robotic navigation, SLAM, 3D reconstruction, and multi-sensor fusion. Traditional registration pipelines mainly rely on geometric information, which becomes unreliable in challenging conditions such as low overlap, repetitive structures, sparse point clouds, or sensor noise.

Recent advances in AI-driven registration attempt to overcome these limitations by integrating semantic information into the matching process. Semantic cues provide high-level contextual understanding of the scene and help distinguish meaningful structures from ambiguous geometric patterns. This trend naturally connects multiple research areas, including 3D geometry, LiDAR sensing, camera-based perception, object detection, semantic segmentation, and machine learning.

In this context, this work focuses on the critical analysis of ML-SemReg, a semantic-aware point cloud registration framework that introduces multi-level semantic consistency to improve correspondence quality. The goal of this report is not only to understand the proposed method, but also to analyze its limitations by identifying realistic failure scenarios and evaluating how these conditions degrade its performance.

2. Methodology and Article Selection Process

2.1 Databases Consulted

IEEE Xplore , SpringerLink , Elsevier / ScienceDirect , Sensors (MDPI) , arXiv .

2.2 Keywords Used

- AI-driven registration , Multi-sensor calibration , Camera pose estimation ,Depth fusion ,Geometric consistency ,Dynamic calibration , Reprojection , Multi-sensor misalignment (RGB + depth + IMU) , Inconsistent calibration .

2.3 Selection Criteria

- Point cloud registration is a fundamental problem in computer vision, serving as a core component of 3D reconstruction, SLAM, robotics, and autonomous navigation systems.

- AI-driven registration represents an intersection between machine learning, image/point cloud segmentation, and geometric vision, allowing high-level semantic information to guide low-level spatial alignment.

- AI-based registration allows researchers to observe, test, and measure improvements on real datasets, ensuring that research outcomes are experimentally grounded.

- This research direction is directly connected to real-world applications, where models are evaluated on real sensor data and performance can be validated in practical deployment scenarios.

3. Study of the Selected Article

Yan, S., Shi, P., Li, J. “ML-SemReg: Boosting Point Cloud Registration with Multi-level Semantic Consistency.”School of Remote Sensing and Information Engineering, Wuhan University; School of Computer Science, Wuhan University. arXiv preprint, 2024.

3.1 Problem Addressed

Correspondence-based point cloud registration (PCR) heavily relies on local geometric descriptors. However, in real-world scenarios (especially low overlap scenes), these methods fail due to:

- Local geometric ambiguity
- Many points share similar local shapes (e.g., corners, planes).
- Leads to a very low correct matching probability.

Two distinct types of mismatches :

- Inter-class mismatching : Matching points belonging to different semantic categories (e.g., car - pole).

- Intra-class mismatching : Points have the same semantic label, but are located in different scene contexts.

Why Existing Methods Fail ?

- Geometry-only descriptors cannot disambiguate similar shapes.

Semantic-assisted methods:

- Usually hard-filter by label, causing loss of true inliers when semantic segmentation is noisy.

- Ignore scene context, thus cannot solve intra-class mismatching.

There is no robust mechanism to simultaneously handle semantic noise, inter-class, and intra-class mismatches, especially in low-overlap point clouds.

3.2 objective of the research

The main objective is to Improve the quality of point correspondences for point cloud registration by explicitly exploiting semantic information at multiple levels, without being fragile to semantic noise.

The authors aim to:

- Reduce inter-class mismatching without hard semantic filtering.

- Suppress intra-class mismatching by incorporating scene-level semantic context.

- Design a plug-and-play framework p that can be integrated into existing registration pipelines without modifying the backend optimization.

3.3 Proposed solution in ML-Semreg

ML-SemReg introduces a semantic-aware correspondence filtering framework based on multi-level semantic consistency. The key idea is to use semantic labels to guide the matching process before geometric registration : filtered correspondences are passed to a classical registration algorithm .

- Input: Two point clouds with associated semantic labels.

- **Initial Correspondence Generation:** Candidate correspondences are generated using a base matching method.

- **Group Matching (Local Semantic Consistency):** Correspondences are grouped by semantic class , inter-class mismatches are filtered out.

- **Mask Matching (Scene Semantic Consistency):** Scene-level semantic masks are applied and Intra-class mismatches are suppressed.

- Output: High-inlier-ratio correspondences.

3.4 Proposed Failure & Weakness scenarios

Scenario	How the Scenario Appears in Real Life	How It Causes Failure in ML-SemReg
Semantic Label Noise	- LiDAR semantic segmentation is performed by a neural network - Objects with similar geometry are confused by the model - Distant objects contain fewer points and are often misclassified - Partial occlusions produce incomplete object shapes	- ML-SemReg groups points based on semantic labels - Wrong labels create incorrect semantic groups - Correct correspondences may be rejected early - Inlier ratio is reduced
Dynamic Objects	- Vehicles move between the acquisition of two scenes - Pedestrians may enter or leave the scene - Objects change position between consecutive LiDAR scans	- ML-SemReg assumes semantic consistency across scans - Moving objects violate this assumption - Correspondences on dynamic objects are unstable - Matching becomes biased or incomplete
LiDAR Motion Distortion (Rolling Acquisition)	- LiDAR scans are acquired sequentially over time - The platform moves during the scan (vehicle motion) - Points are captured at slightly different poses	- ML-SemReg assumes rigid point clouds - Local neighborhoods are geometrically distorted - Semantic structure descriptors become inconsistent - Correspondence quality decreases
LiDAR Pose Drift Between Scenes	- LiDAR sensor slightly shifts its position or orientation - Mechanical vibrations affect sensor mounting - Temperature changes cause material expansion - Sensor pose differs between two scene acquisitions	- ML-SemReg assumes stable sensor pose between scenes - Semantic neighborhoods are misaligned - Guiding semantic structures do not correspond - Registration accuracy degrades

4. Experiments and results

4.1 Metrics

- **Inliers**: Number of correct correspondences found between the two point clouds.
- **Inlier Ratio (IR)**: Percentage of correct correspondences out of all proposed correspondences.
- **Rotation Error (RRE)**: Angular difference (in degrees) between the estimated rotation and the ground truth rotation.
- **Translation Error (RTE)**: Positional difference (in cm or meters) between the estimated translation and the ground truth translation.
- **Success**: Indicates whether the registration is considered successful according to predefined thresholds for RRE and RTE.

4.2 Scenario 1 results

In this failure scenario, semantic labels are artificially corrupted before being fed to ML-SemReg. Specifically, a fraction of points are randomly mislabeled. ML-SemReg heavily relies on semantic labels: the Group Matching (GM) step eliminates cross-class correspondences, and Mask Matching (MM) leverages local semantic context. When labels are incorrect, correct matches are removed while incorrect matches survive, causing significant registration drift. The experimental results confirm this behavior: most frames exhibit extremely low inlier ratios (1–3%) and very high rotation (170) and translation errors (10m), with only a few frames achieving successful registration. The graphs of inlier ratios and errors illustrate that semantic label corruption directly leads to registration failures, highlighting the critical importance of accurate semantic segmentation for robust performance.

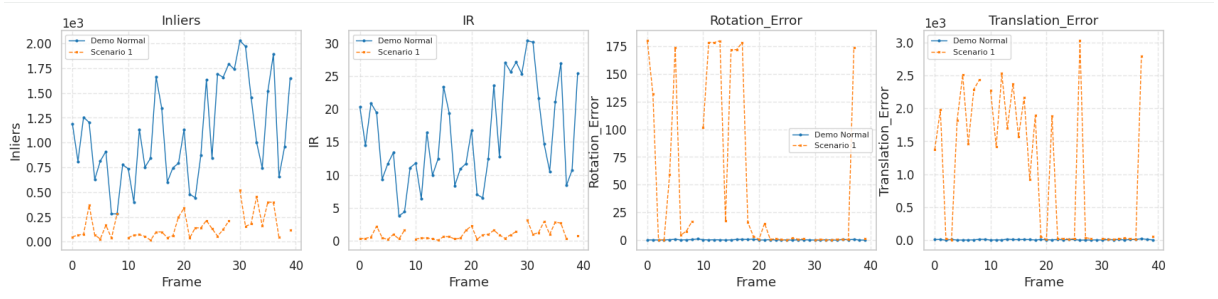


Figure 1: Comparison of Demo Normal vs Scenario 1 – Impact of wrong labeling Objects on the performance.

4.3 Scenario 2 results

In this failure scenario, dynamic objects are removed from only one scan to simulate real-world motion, where objects like cars, pedestrians, or bicycles appear in one scan but are absent in the other. ML-SemReg relies on semantic consistency: Group Matching (GM) expects corresponding semantic classes between scans, and Mask Matching (MM) depends on similar local semantic neighborhoods. When dynamic objects are missing in one scan, GM may discard valid keypoints or attempt incorrect matches, and MM fails due to asymmetric scene neighborhoods. The experimental results confirm this behavior: several frames fail registration completely (GSS failed no anchors), while frames that succeed exhibit lower inlier ratios (4–27%) and higher rotation (up to 1) and translation errors (2–30cm) compared to normal registration. These results demonstrate that the presence of moving objects, or their absence in one scan, degrades registration performance, highlighting ML-SemReg’s sensitivity to dynamic changes in the scene.

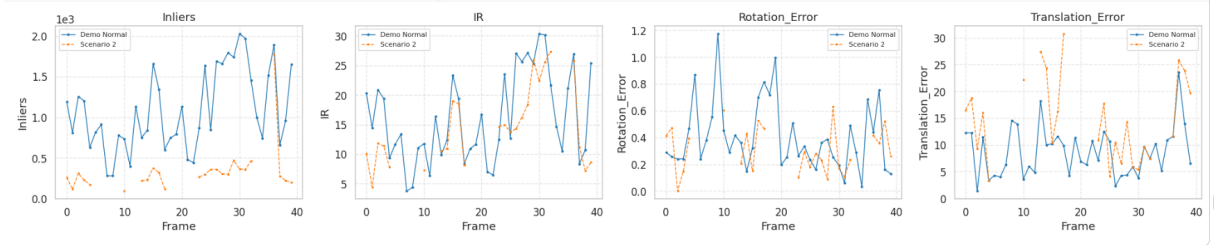


Figure 2: : Comparison of Demo Normal vs Scenario 2 – Performance After Removal of Dynamic Objects.

4.4 Scenario 3 results

In this failure scenario, point clouds are distorted to simulate LiDAR motion during scanning, also known as rolling shutter effects. Specifically, the Y coordinates of points are shifted according to their X positions using a sinusoidal function, creating a subtle “bend” in the scene. This simulates the fact that a LiDAR does not capture all points at once: each point is measured at a slightly different instant, and small movements of the LiDAR or vehicle cause neighboring points to be displaced relative to each other. ML-SemReg relies on local neighborhoods of points to compute Neighborhood Semantic Signatures (NSS) and Guiding Semantic Structures (GSS). When neighborhoods are distorted, these structures no longer correctly represent the scene geometry, which can reduce the number of correct matches during registration.

The experimental results confirm this behavior: while all frames are technically successful, inlier ratios vary widely (3–30%), and rotation and translation errors increase compared to the normal scenario, with RRE up to 1.5° and RTE up to 23cm in some frames. These results illustrate that even subtle distortions of local point neighborhoods

without changing individual point measurements can degrade registration performance, highlighting the sensitivity of ML-SemReg to geometric inconsistencies in point cloud data.

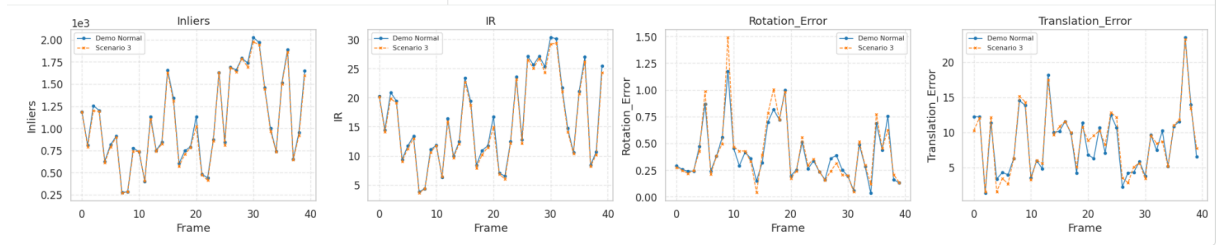


Figure 3: Comparison of Demo Normal vs Scenario 3 – Effect of Rolling Shutter Distortion on Feature Matching and Pose Estimation.

4.5 Scenario 4 results

In this scenario, small errors in the LiDAR extrinsic calibration are simulated by slightly rotating (2° around Y) and translating (0.1m along X) the target scan. Such drifts can occur in practice due to vibrations, temperature changes, or mount loosening.

ML-SemReg relies on accurate neighborhood relationships and semantic structures for matching. Even minor mis-calibrations shift keypoints and corrupt these structures, reducing matching accuracy.

Experimental results show very low inlier ratios (1–17%) and high rotation/translation errors (RRE $1.5\text{--}2.6^\circ$, RTE 33–83cm), with almost all frames failing. This contrasts with normal calibration, where inlier ratios are much higher (3–30%) and all frames succeed.

This highlights that precise LiDAR calibration is crucial for robust ML-SemReg performance.

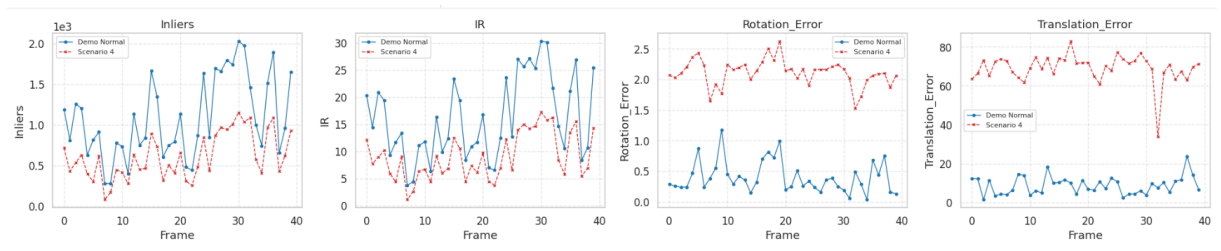


Figure 4: Comparison of Demo Normal vs Scenario 4 – Effect of Calibration Drift on Inliers, Pose Accuracy, and Registration Success.

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